

Counter-Swarm Defence

UI/UX screen compositions and simulation lab views for architecture review.
These are design specifications rendered for PDF review — not production software.

Author: Victor.I · Screens: X01, X03, X04, X05, I01, S01 · Simulation loop included
Source markdown: docs/product/* · docs/simulation/*

COVERAGE REDUCED — RF-02 offline · track confidence capped in hatched sector

COUNTER-SWARM · Site ALPHA · Mode MONITOR · op_j.smith

SYS OK

SENS 3/4

LAG 0.12s

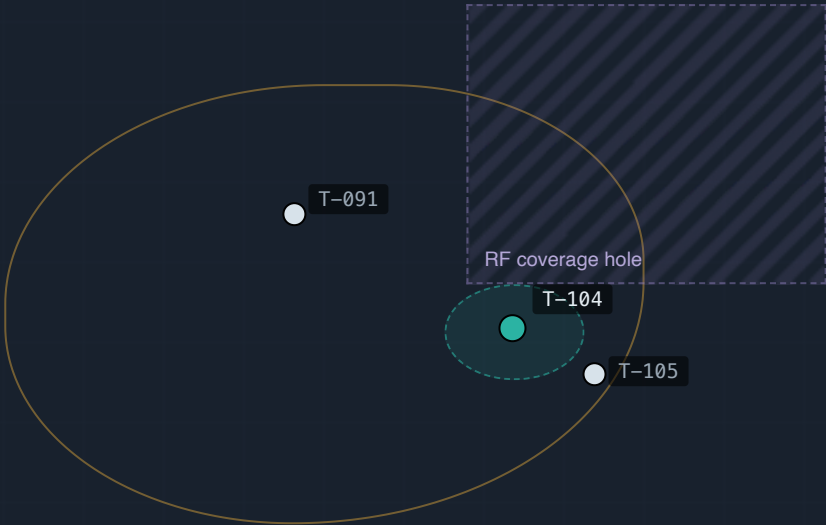
ALERTS

T3	Swarm indicators	2s
T2	New track T-104	8s
T1	RF-02 heartbeat	42s

ACTIVE TRACKS

T-104 UAV? 0.72	GOOD
T-105 UAV? 0.68	FAIR
T-091 Bird? 0.61	GOOD
T-088 Unk 0.40	COAST

MAP · layers on: tracks, uncertainty, geofence



Tracks

Uncertainty

FOV

Obs

Predict

Simplify

RISK / NEXT ACTION

Track T-104

Risk: ELEVATED · policy v3

INFERENCE

Class UAV p=0.72 · det-v3.2

RECOMMENDATION

HEIGHTEN_MONITORING (P-12)

MISSING INFO

RF corroboration

Evidence

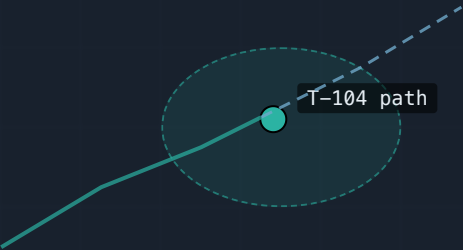
Decide

TIMELINE · Sensors · System · Audit · Assistant

12:01:02 obs RDR-1 · 12:01:03 det · 12:01:04 track upd · 12:01:05 risk elevated · 12:01:06 alert T2



FOCUSED MAP · other tracks dimmed



Evidence pack

Observation ≠ inference ≠ prediction ≠ recommendation ≠ decision

OBSERVATION

obs-881 RDR-1 · obs-882 EO-3 · quality OK · 3 plots / 8s

INFERENCE

UAV 0.72 · Bird 0.18 · Unknown 0.10 · model det-v3.2

PREDICTION

60s kinematic envelope (dashed on map)

RECOMMENDATION

HEIGHTEN_MONITORING · policy P-12

DECISION

Pending human

Evidence graph: obs → det-9 → track T-104 → behaviour? → risk ELEVATED

Cue sensor

Open incident

Decide

Record authorised response category

Context: Track T-104 · Incident I-22 · Recommended: HEIGHTEN_MONITORING

- DISMISS
- HEIGHTEN_MONITORING
- CUE_SENSOR
- NOTIFY_EXTERNAL
- REQUEST_ESCALATION
- OPEN_INCIDENT

Rationale (required)

Multi-track proximity + shared heading; RF still missing on T-109...

This records intent for authorised systems. It does not control weapons or electronic attack.

Cancel

Record decision

INCIDENT I-22 · SUSPECTED_COORDINATION · owner cmdr.lee · ACTIVE

RISK HIGH BAND

3 tracks

INCIDENT MAP · members only



SUMMARY

BEHAVIOUR INDICATOR
Formation-like proximity (not a confirmed swarm fact)

MISSING
RF on T-109

RECOMMENDATION
NOTIFY_EXTERNAL

Decide

Transfer

Close

HANDOFF

Status: NOT_REQUIRED until decision

SHARED TIMELINE · Notes · Participants

12:02 incident opened · 12:03 behaviour flag · 12:04 risk high · awaiting decision

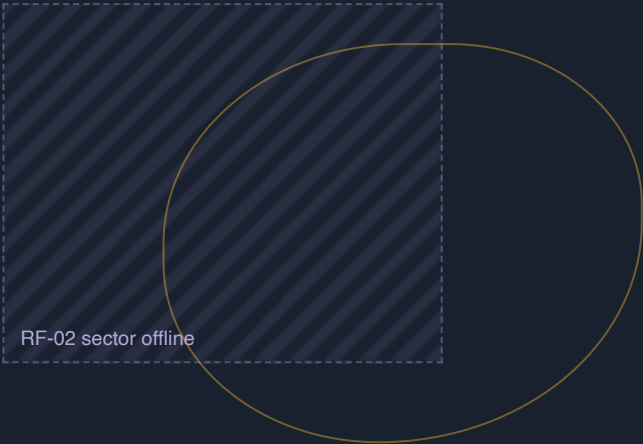
Sensors

ID	Type	Status	Last obs	Notes
RDR-1	radar	OK	0.4s	—
EO-3	eo	OK	1.1s	—
RF-02	rf	DOWN	42s	heartbeat miss
MIC-1	acoustic	OK	0.8s	—

Platform

Component	Status	Signal
API / Realtime	OK	p95 180ms
Event bus lag	OK	120ms
Tracking	OK	4 active
Model det-v3.2	OK	infer p95 42ms

COVERAGE MAP



Simulation → real software path

No weapons, no jamming, no kinetic effects — categories and mock externals only.



Scenario	ID	What you see on X01
Coordinated trio	SCN-SWM-01	T3 behaviour alert + incident offer
RF silent	SCN-MOD-01	Missing-info + confidence ceiling
Sensor death	SCN-FLT-01	Coverage hatch + X05 DOWN
Handoff fail	SCN-DEC-02	Decision saved · HANDOFF_FAILED

Scenario control

Catalogue → run → inject faults → observe operator console

Field	Value
Scenario	SCN-SWM-01 coordinated_trio_v3
Seed	42
Speed	1x
Elapsed	00:01:14 / 00:05:00

Start

Pause

Inject fault

FAULT TOGGLES

- ☐ Drop RF
- ☐ Delay EO
- ☐ Duplicate
- ☐ Corrupt
- ☒ Sensor death RF-02 @ t=45s
- ☐ Link loss

TRUTH OVERLAY

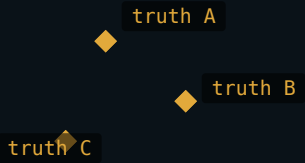
- Instructor only

Off

Forbidden for operator-under-test

Mock external will ACK/FAIL category handoffs. No physical effects exist in this environment.

TRUTH VIEW (instructor)



Diamonds = truth objects · Operator X01 sees estimated tracks only · Scorer compares offline

S01 DIRECTOR

LAB

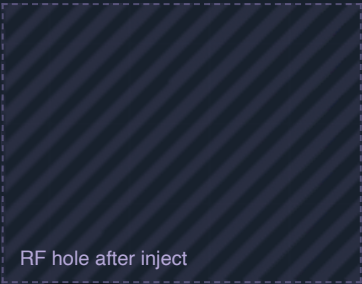
Running SCN-SWM-01
Fault: RF-02 death armed
Obs rate: 42/s → bus
Mock external: idle

Inject now

X01 OPERATOR

SENS 3/4

Same real console UI · fed by simulation



RF hole after inject

T-104



Related markdown (source of truth for review comments)

Topic	Path
Wireframes	docs/product/04-wireframes.md
Screen inventory	docs/product/05-ui-screen-inventory.md
UX principles / tokens	docs/product/06-*.md · 08-design-tokens.md
Workflows / journeys	docs/product/01-*.md · 02-*.md · 10-*.md
Simulation pack	docs/simulation/*
Digital twin architecture	docs/systems/digital-twin.md

PDFs in docs/product/pdfs/ are generated from this HTML for stakeholder review. High-fidelity design comps can replace these later without changing screen IDs.